# Tutorial 1

1. Simplify the following complex numbers

   (a) $(3-4\mathrm{i})(6+2\mathrm{i})$.

   (b) $\dfrac{2+\mathrm{i}}{4-7\mathrm{i}}$.

   (c) $(17-6\mathrm{i})(-4-12\mathrm{i})^*$.

   (d) $\mathrm{i}^3-4\mathrm{i}^2+2$.

   (e) $\left(\dfrac{-6+2\mathrm{i}}{1-8\mathrm{i}}\right)^2$.

2. Determine the magnitudes and arguments of the following complex numbers.

   (a) $3\mathrm{i}$.

   (b) $-3+2\mathrm{i}$.

   (c) $-4$.

3. Prove that, for any positive integer $n$,

$$\mathrm{i}^{4n}=1, \quad \mathrm{i}^{4n+1}=\mathrm{i}, \quad \mathrm{i}^{4n+2}=-1, \quad \text{and} \quad \mathrm{i}^{4n+3}=-\mathrm{i}.$$

4. Show that

   (a) $\mathrm{Re}(\mathrm{i}z)=-\mathrm{Im}(z)$.

   (b) $\mathrm{Im}(\mathrm{i}z)=\mathrm{Re}(z)$.

5. Reduce each of these quantities to a real number

   (a) $\dfrac{1+2\mathrm{i}}{3-4\mathrm{i}}+\dfrac{2-\mathrm{i}}{5\mathrm{i}}$.

   (b) $(1-\mathrm{i})^4$

6. Let $z=a+\mathrm{i}b$. Determine $\mathrm{Re}(z^2)$ and $\mathrm{Im}(z^2)$.

7. Show that complex numbers $z$, $w$, and $u$ form vertices of an equilateral triangle if

$$z^2+w^2+u^2=zw+zu+wu. \tag{1}$$

8. Show that $z^2=(z^*)^2$ if $z$ is either purely real or purely imaginary.

9. Let $z$ and $w$ be numbers with $z^*w\neq 1$. Suppose either $z$ or $w$ has magnitude 1. Prove that

$$\left|\frac{z-w}{1-z^*w}\right|=1.$$

10. Show that, for any numbers $z$ and $w$,

$$|z+w|^2+|z-w|^2=2\left(|z|^2+|w|^2\right).$$

11. In the following, find $u$ and $v$ so that $f = u + \mathrm{i}v$, determine all the points where the Cauchy–Riemann equations hold, and determine all $z$ at which $f$ is differentiable. Familiar facts about continuity of real-valued functions of two real variables can be assumed.

    (a) $f(z) = z - \mathrm{i}$. (b) $f(z) = |z|$. (c) $f(z) = \mathrm{i}|z|^2$.

    (d) $f(z) = \dfrac{z}{\mathrm{Re}(z)}$. (e) $f(z) = (z^*)^2$.

12. Write the following function value in the form $a + \mathrm{i}b$.

    (a) $\mathrm{e}^{\mathrm{i}}$. (b) $\cos(3 + 2\mathrm{i})$. (c) $\mathrm{e}^{5+2\mathrm{i}}$.

    (d) $\sin^2(1 + \mathrm{i})$. (e) $\mathrm{e}^{\mathrm{i}\pi/2}$.

13. Determine $u$ and $v$ such that

$$\mathrm{e}^{z^2} = u(x, y) + \mathrm{i}v(x, y). \tag{2}$$

    Show that $u$ and $v$ satisfy the Cauchy–Riemann equations for all $(x, y)$.

14. Find the derivative of the following functions

    (a) $\frac{1}{z}$. (b) $\mathrm{e}^{z^2}$.

15. Determine the logarithms of the following functions

    (a) $-4\mathrm{i}$. (b) $-5$. (c) $-9 + 2\mathrm{i}$.

16. Evaluate the following complex powers.

    (a) $\mathrm{i}^{1+\mathrm{i}}$. (b) $\mathrm{i}^{\mathrm{i}}$. (c) $(-1 + \mathrm{i})^{-3\mathrm{i}}$.

    (d) $\mathrm{i}^{1/4}$. (e) $(-4)^{2-\mathrm{i}}$. (f) $1^{1/6}$.